Major Rudolf Anderson's U-2F had overflown Guantánamo Bay and turned north and east when the Soviet 507th Anti-Aircraft Missile Regiment launched a salvo of three missiles. The first two missed. The third showered the plane with shrapnel, killing the pilot. During the crisis, several servicemen died in accidents, including the crew of a US RB-47 and several members of the Meteor FKR regiment en route to their firing position near Guantánamo. Major Anderson became the only combat fatality of the crisis (and the first recipient of the Air Force Cross).

The decision to fire the surface-to-air missiles was not taken by General Pliyev, but by his subordinates. When the local commander sought permission to shoot down the aircraft, General Pliyev was out of contact, so his deputy, Lieutenant General Grechko, and the air defence chief, Lieutenant General Garbuz, took responsibility. Fearing an invasion was impending, they assumed the reconnaissance aircraft augured an imminent attack.

US signals intelligence learned from Cuban transmissions that the plane had been shot down and Major Anderson killed. ExComm was told around 4.00pm, as it was grappling with a diplomatic response to Khrushchev's letters. There was immediate speculation that the act was coordinated with a hardening of the Soviet line, and perhaps indicative of a hard-line faction in the Kremlin pushing Khrushchev toward confrontation. The possibility that the missiles had been fired by Cuban soldiers added to the conjecture. There emerged an implausible suggestion that Fidel Castro had personally fired the weapon.

In Moscow, the news was an unwelcome shock to Nikita Khrushchev. He was concerned that he might be losing control of the military at a critical moment in the search for a diplomatic way out of the crisis. Such was Khrushchev's grasp of Soviet command and control, he believed that Castro had indeed ordered the missles to be fired, and sent him a stern rebuke. The mildest of rebukes was issued to the Soviets officers. Marshal Malinovsky cabled Pliyev that with a possible diplomatic resolution emerging, 'we believe you were too hasty'. Pliyev's reaction to his subordinates' action is unknown.